

# Curriculum Vitae

Takashi Furuya

## Takashi Furuya

I am an Assistant Professor in the Faculty of Life and Medical Sciences at Doshisha University and a Visiting Scientist at the RIKEN Center for Advanced Intelligence Project.

My research interests lie at the intersection of applied mathematics and machine learning, with a particular focus on:

- Neural operators and operator learning
- Transformers and in-context learning
- Approximation theory for deep learning
- Inverse problems for partial differential equations

## Contact Information

E-mail: [tfuruya@mail.doshisha.ac.jp](mailto:tfuruya@mail.doshisha.ac.jp)

Personal: [takashi.furuya0101@gmail.com](mailto:takashi.furuya0101@gmail.com)

## Education and Professional Experience

- **September 2025–present:** Visiting Scientist, RIKEN Center for Advanced Intelligence Project, Computational Physics Machine Learning Team, Japan
- **April 2025–present:** Assistant Professor, Faculty of Life and Medical Sciences, Doshisha University, Japan
- **August 2022–March 2025:** Assistant Professor, Education and Research Center for Mathematical and Data Science, Shimane University, Japan
- **April 2021–July 2022:** JSPS Postdoctoral Fellow, Department of Mathematics, Hokkaido University, Japan
- **2018–2021:** Ph.D. in Mathematics, Graduate School of Mathematics, Nagoya University
- **2016–2018:** M.Sc. in Mathematics, Graduate School of Mathematics, Nagoya University
- **2012–2016:** B.Sc. in Mathematics, College of Mathematics, University of Tsukuba

## Publications

### Under Review

1. **T. Furuya**, D. Mis, I. Dokmanić, M. V. de Hoop, M. Lassas.  
*Function Graph Transformers Universally Approximate Operators between Function Spaces.*  
arXiv:2605.17968 (2026).
2. R. Barboni, **T. Furuya**, M. V. de Hoop, G. Peyré.  
*Training Infinitely Deep and Wide Transformers.*  
arXiv:2605.17660 (2026).
3. **T. Furuya**, Y. Korolev, T. Yaguchi.  
*Approximation of Maximally Monotone Operators: A Graph Convergence Perspective.*  
arXiv:2605.12301 (2026).
4. **T. Furuya**, R. Ozawa, Jenn-Nan Wang.  
*Approximation Theory of Laplacian-Based Neural Operators for Reaction-Diffusion System.*  
arXiv:2605.12025 (2026).

5. **T. Furuya**, A. Kratsios, D. Possamai, B. Raonić.  
*One Model to Solve Them All: 2BSDE Families via Neural Operators.*  
arXiv:2511.01125 (2025).
  6. **T. Furuya**, M. V. de Hoop, M. Lassas.  
*Transformers through the Lens of Support-Preserving Maps between Measures.*  
arXiv:2509.25611 (2025).
  7. A. Kratsios, **T. Furuya**.  
*Is In-Context Universality Enough? MLPs Are Also Universal In-Context.*  
arXiv:2502.03327 (2025).
  8. A. Kratsios, **T. Furuya**.  
*Simultaneously Solving FBSDEs with Neural Operators of Logarithmic Depth, Constant Width, and Sublinear Rank.*  
arXiv:2410.14788 (2024).
  9. A. Kratsios, **T. Furuya**, L. B. J. Antonio, M. Lassas, M. V. de Hoop.  
*Mixture of Experts Soften the Curse of Dimensionality in Operator Learning.*  
arXiv:2404.09101 (2024).
  10. **T. Furuya**, H. Kusumoto, K. Taniguchi, N. Kanno, K. Suetake.  
*Theoretical Error Analysis of Entropy Approximation for Gaussian Mixtures.*  
arXiv:2202.13059 (2022).
- 

## Publications

1. **T. Furuya**, D. Murari, C. B. Schönlieb.  
*Approximation Theory for Lipschitz Continuous Transformers.*  
**International Conference on Machine Learning (ICML)**, 2026.
2. A. Kratsios, B. J. Kim, **T. Furuya**.  
*Approximation Rates in Besov Norms and Sample-Complexity of Kolmogorov-Arnold Networks with Residual Connections.*  
**Neural Networks**, **200**, 108797 (2026).
3. **T. Furuya**, P. Z. Kow, J. N. Wang.  
*Consistency of the Bayes Method for the Inverse Scattering Problem with Randomly Truncated Sieve Priors.*  
**Inverse Problems and Imaging**, **21**, 222–244 (2026).
4. D. Murari, **T. Furuya**, C. B. Schönlieb.  
*Approximation Theory for 1-Lipschitz ResNets.*  
**Advances in Neural Information Processing Systems (NeurIPS)**, **38**, 142888–142919 (2025).
5. **T. Furuya**, M. V. de Hoop, G. Peyré.  
*Transformers Are Universal In-Context Learners.*  
**International Conference on Learning Representations (ICLR)**, 80820–80845 (2025).
6. **T. Furuya**, K. Taniguchi, S. Okuda.  
*Quantitative Approximation for Neural Operators in Nonlinear Parabolic Equations.*  
**International Conference on Learning Representations (ICLR)**, 85528–85556 (2025).
7. **T. Furuya**, J. N. Wang.  
*The Bernstein–von Mises Theorem for the Inverse Scattering Problem.*  
**Communications in Mathematical Sciences**, **23**(4), 1023–1042 (2025).
8. H. S. O. Borde, **T. Furuya**, A. Kratsios, M. T. Law.  
*Approximation Rates and VC-Dimension Bounds for (P)ReLU MLP Mixture of Experts.*  
**Transactions on Machine Learning Research** (2025).
9. **T. Furuya**, M. Puthawala, M. Lassas, M. V. de Hoop.  
*Can Neural Operators Always Be Continuously Discretized?*  
**Advances in Neural Information Processing Systems (NeurIPS)**, **37**, 98936–98993 (2024).

10. **T. Furuya**, S. Okuda, K. Suetake, Y. Sawada.  
*Convergences for Minimax Optimization Problems over Infinite-Dimensional Spaces Towards Stability in Adversarial Training.*  
**Transactions on Machine Learning Research** (2024).
11. L. B. J. Antonio, **T. Furuya**, F. Faucher, A. Kratsios, X. Tricoche, M. V. de Hoop.  
*Out-of-Distribution Risk Bounds for Neural Operators with Applications to the Helmholtz Equation.*  
**Journal of Computational Physics**, **513**, 113168 (2024).
12. **T. Furuya**, P. Z. Kow, J. N. Wang.  
*Consistency of the Bayes Method for the Inverse Scattering Problem.*  
**Inverse Problems**, **40**, 055001 (2024).
13. **T. Furuya**, M. Puthawala, M. Lassas, M. V. de Hoop.  
*Globally Injective and Bijective Neural Operators.*  
**Advances in Neural Information Processing Systems (NeurIPS)**, **36**, 57713–57753 (2023).
14. M. V. de Hoop, **T. Furuya**, C. L. Lin, G. Nakamura, M. Vashisth.  
*Local Recovery of a Piecewise Constant Anisotropic Conductivity in EIT on Domains with Exposed Corners.*  
**Inverse Problems**, **39**, 025005 (2023).
15. **T. Furuya**, R. Potthast.  
*Inverse Medium Scattering Problems with Kalman Filter Techniques.*  
**Inverse Problems**, **38**, 095003 (2022).
16. **T. Furuya**, K. Suetake, K. Taniguchi, H. Kusumoto, R. Saiin, T. Daimon.  
*Spectral Pruning for Recurrent Neural Networks.*  
**Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)**, PMLR Vol. 151 (2022).
17. **T. Furuya**.  
*Remarks on the Factorization and Monotonicity Method for Inverse Acoustic Scattering.*  
**Inverse Problems**, **37**, 065006 (2021).
18. **T. Furuya**.  
*The Factorization and Monotonicity Method for the Defect in an Open Periodic Waveguide.*  
**Inverse and Ill-Posed Problems**, **28**, 783–796 (2020).
19. **T. Furuya**.  
*Scattering by the Local Perturbation of an Open Periodic Waveguide in the Half Plane.*  
**Journal of Mathematical Analysis and Applications**, **489**, 124149 (2020).
20. **T. Furuya**.  
*The Direct and Inverse Scattering Problem for the Semilinear Schrödinger Equation.*  
**NoDEA Nonlinear Differential Equations and Applications**, **27**, Article 24 (2020).
21. T. Daimon, **T. Furuya**, R. Saiin.  
*The Monotonicity Method for the Inverse Crack Scattering Problem.*  
**Inverse Problems in Science and Engineering**, **28**, 1570–1581 (2020).
22. **T. Furuya**.  
*A Modification of the Factorization Method for Scatterers with Different Physical Properties.*  
**Mathematical Methods in the Applied Sciences**, **42**, 4017–4030 (2019).
23. H. Chihara, **T. Furuya**, T. Koshikawa.  
*Hermite Expansions of Some Tempered Distributions.*  
**Journal of Pseudo-Differential Operators and Applications**, **9**, 105–124 (2017).

## Talks and Presentations

- 2014-03-01. **Conference Proceeding talk 3 on Relevant Topic in Your Field.** Testing Institute of America 2014 Annual Conference. Los Angeles, CA, USA
- 2014-02-01. **Talk 2 on Relevant Topic in Your Field.** London School of Testing. London, UK
- 2013-03-01. **Tutorial 1 on Relevant Topic in Your Field.** UC-Berkeley Institute for Testing Science. Berkeley, CA, USA

- 2012-03-01. **Talk 1 on Relevant Topic in Your Field.** UC San Francisco, Department of Testing. San Francisco, CA, USA

## Funding and Fellowships

### Competitive Research Funding

*PI: Principal Investigator; Co-I: Co-Investigator.*

#### Royal Society ISPF International Collaboration Awards 2025

##### Co-Investigator (Co-I)

**Project:** *Operator learning for gradient flows with applications in imaging*

**Principal Investigator:** Yury Korolev

**Grant No.:** ICA/R2/252033

**Period:** March 2026 – March 2029

#### JSPS KAKENHI Grant-in-Aid for Early-Career Scientists

##### Principal Investigator (PI)

**Project:** *A study of operator learning for inverse problems in partial differential equations*

**Grant No.:** 24K16949

**Period:** April 2024 – March 2027

**Amount:** JPY 3,900,000

[Project page](#)

#### Sumitomo Foundation Grant for Basic Science Research 2023

##### Principal Investigator (PI)

**Project:** *Approximate solutions for ill-posed inverse problems of partial differential equations using deep learning*

**Period:** December 2023 – November 2024

**Amount:** JPY 1,000,000

[Project page](#)

#### Yamanashi Prefectural Grant for Young Researchers

##### Principal Investigator (PI)

**Project:** *Reconstruction methods for inverse problems in partial differential equations using deep learning*

**Period:** August 2022 – March 2023

**Amount:** JPY 1,000,000

[Project page](#)

## International Research Collaboration

### JSPS–NSF International Research Exchange Program

**Host researcher:** Prof. Michael A. Puthawala (South Dakota State University)

**Host institution:** Doshisha University, Japan

**Period:** January – February 2026

## Fellowships

### JSPS Postdoctoral Fellowship (PD)

**Grant No.:** 21J00119

**Institution:** Hokkaido University

**Period:** April 2021 – March 2023

[Project page](#)

### **JSPS Research Fellowship for Young Scientists (DC2)**

**Institution:** Nagoya University

**Period:** April 2019 – March 2021

[Project page](#)

### **Tobitate! Study Abroad JAPAN**

**Host institution:** Karlsruhe Institute of Technology, Germany

**Period:** October 2018 – February 2019

[Program page](#)